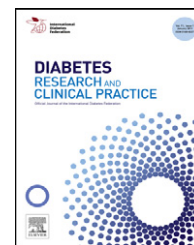


Contents available at [Sciverse ScienceDirect](http://www.sciencedirect.com)

Diabetes Research and Clinical Practice

journal homepage: www.elsevier.com/locate/diabres

International
Diabetes
Federation



Community Diabetes Education (CoDE) for uninsured Mexican Americans: A randomized controlled trial of a culturally tailored diabetes education and management program led by a community health worker[☆]

Elizabeth A. Prezio^{a,*}, Dunlei Cheng^{a,1}, Bijal A. Balasubramanian^{a,1},
Kerem Shuval^{a,1}, Darla E. Kendzor^{a,1}, Dan Culica^{b,2}

^a University of Texas School of Public Health, 6011 Harry Hines Blvd., Suite V8.112, Dallas, TX 75390, United States

^b TMF Health Quality Institute, Bridgepoint I, Suite 300, 5918 West Courtyard Drive, Austin, TX 78730-5036, United States

ARTICLE INFO

Article history:

Received 5 November 2012

Received in revised form

24 December 2012

Accepted 25 January 2013

Published on line 28 February 2013

Keywords:

Community health worker

Diabetes management

Hispanic

Uninsured

ABSTRACT

Aims: The purpose of this randomized controlled trial was to determine the impact of a culturally tailored diabetes education program led by a community health worker (CHW) on the HbA1c, blood pressure, body mass index (BMI) and lipid status of uninsured Mexican Americans with diabetes.

Methods: Adult patients were recruited from a community clinic and randomized into intervention ($n = 90$) and control ($n = 90$) groups. Both groups received usual medical care from clinic physicians. The intervention group participated in the Community Diabetes Education (CoDE) program over 12 months. The primary outcome of interest was HbA1c. Secondary outcomes included blood pressure, BMI and lipid status. Variations in outcomes over time were assessed within groups and between groups using linear mixed-models and an intention-to-treat approach. Assessment of changes in HbA1c, blood pressure and lipid status over 12 months included variables to control for modifications made to antidiabetic, antihypertensive and lipid lowering medications.

Results: There was no difference in baseline characteristics between the intervention and control groups. Mean changes of HbA1c over 12 months showed a significant intervention effect (-0.7% , $p = .02$) in the CoDE group compared with controls. HbA1c decreased significantly from baseline to 12 months within the intervention (-1.6% , $p < .001$) and control (-0.9% , $p < .001$) groups. No differences between groups for secondary outcomes were found. **Conclusions:** This study supports the effectiveness of CHWs as diabetes educators/case managers functioning as integral members of the health care team in community clinic settings serving uninsured Mexican Americans.

© 2013 Elsevier Ireland Ltd. All rights reserved.

[☆] ClinicalTrials.gov Identifier: NCT00151190.

* Corresponding author. Tel.: +1 214 801 9843; fax: +1 512 428 5786.

E-mail addresses: eaprezio@sbcglobal.net (E.A. Prezio), dunlei.cheng@utsouthwestern.edu (D. Cheng), bijal.balasubramanian@utsouthwestern.edu (B.A. Balasubramanian), kerem.shuval@utsouthwestern.edu (K. Shuval), darla.kendzor@utsouthwestern.edu (D.E. Kendzor), dan.culica@tmf.org (D. Culica).

¹ Tel.: +1 214 648 1080.

² Tel.: +1 512 334 1679.

0168-8227/\$ – see front matter © 2013 Elsevier Ireland Ltd. All rights reserved.

<http://dx.doi.org/10.1016/j.diabres.2013.01.027>

1. Introduction

The prevalence and incidence of diabetes mellitus in the United States (US) continue to rise, and diabetes now affects 25.8 million children and adults [1]. In 2010, there were 1.9 million new cases of diabetes in people over the age of 20 [1]. Diabetes affects 11.8% of Hispanics compared with 7.1% of non-Hispanic whites, and among Hispanics, diabetes affects 13.3% of Mexican Americans [1]. While the Hispanic population accounts for 16% of the US populace, 19% of US Hispanics live in Texas, constituting 38% of the state residents [2]. Furthermore, Mexican Americans, the largest Hispanic/Latino subgroup, are 1.9 times more likely to have diabetes and 1.6 times more likely to develop the diabetes complication of end stage renal disease than non-Hispanic whites of similar age [3,4].

Adequate glycemic control reduces the likelihood of developing long-term complications from diabetes including retinopathy, nephropathy and neuropathy [5,6]. Diabetes self-management education (DSME) is essential to increase the knowledge and skills of patients with diabetes in order to achieve this goal [7]. The importance of culturally tailored DSME was established in 2002 by a landmark intervention designed for Mexican Americans in Starr County, Texas [8]. Since that time multiple culturally tailored diabetes education interventions have been developed and implemented in Hispanic/Latino populations with documented improvements in knowledge, behaviors and clinical outcomes [8–19]. A meta-analysis examining the effect of culturally tailored diabetes education in ethnic minorities showed that these interventions were effective for improving glycemic control, though the magnitude of effect was greater in hospital- or clinic-based settings than in community settings [20]. In addition, greater improvements were observed in patients with baseline hemoglobin A1c (HbA1c) levels $\leq 8.5\%$, and maximum improvement in HbA1c occurred at 6 months followed by a decline in glycemic control [20].

In response to the ever increasing demand for diabetes care services in socially disadvantaged populations, community health workers (CHWs) have emerged as an available resource to reduce disparities in health outcomes [21]. CHWs are typically bilingual and bicultural members of the local community who serve as bridges between patients and health care providers in underserved areas [22]. Systematic reviews of the effectiveness of CHWs in the care of patients with diabetes have documented improvements in health outcomes for vulnerable populations, including Mexican Americans [9,21–25]. A role for CHWs in diabetes care is supported by the American Association of Diabetes Educators and the American Public Health Association, but the work of CHWs has not found relevance within the Chronic Care Model of evidence based practices [26–28]. Specific roles for CHWs in diabetes management have not been firmly established because few studies employing rigorous methodology have utilized CHWs in positions bearing full responsibility for delivery of DSME interventions to patients [10,16,19]. In addition, published studies have often failed to include detailed information about program development and implementation along with outcomes and measures of association between interventions

and outcomes [29]. A systematic review of community-based diabetes education interventions targeting disadvantaged populations concluded that successful interventions were culturally tailored, individualized, led by community educators, of sufficient intensity (>10 contact times) and delivered over a long duration (≥ 6 months) [23].

Of patients with diabetes who have some form of health insurance, 56% report ever attending DSME, while only 42% of patients who are uninsured ever attend DSME [30,31]. As the numbers of Certified Diabetes Educators (CDEs), diabetes nurse educators, and DSME programs are limited, the growing number of uninsured individuals in the US has forced communities and health care agencies to address the limited availability of quality DSME programs by engaging alternative lower cost personnel such as CHWs to provide diabetes education services. Furthermore, the number of culturally tailored diabetes education interventions involving large numbers of uninsured patients is limited, and to our knowledge no randomized controlled trials of culturally tailored diabetes interventions have been published involving exclusively uninsured patient populations [8,32].

To date, few randomized controlled trials have been completed which evaluate the effectiveness of CHWs as integral members of clinical care teams with defined roles and responsibilities as patient educators in community settings [8,10,15–19,33]. Even fewer studies have utilized CHWs as the sole provider of educational services without any direct involvement with more qualified personnel such as CDEs and nurse case managers [10,13,16,19]. The Community Diabetes Education (CoDE) program was developed to provide access to DSME for uninsured patients with diabetes in Dallas, Texas where high diabetes prevalence (11.7%) is coupled with a high percentage of residents without health insurance (36.9%) [1,13,34,35]. We report the results of a randomized controlled trial that tested whether a culturally tailored diabetes education and management program (CoDE) led solely by a CHW improved HbA1c levels, blood pressure, lipid levels and BMI more than usual care alone among uninsured Mexican Americans.

2. Methods

2.1. Design

A prospective, randomized controlled repeated measures design was employed to compare the intervention effects between: (1) an intervention group that received a culturally tailored diabetes education and management program (CoDE) along with usual medical care; and (2) a wait-listed comparison group that received only usual medical care. All patients were given informed consent in the preferred language of the study subject followed by (1:1) assignment to either the intervention or control groups using a computer generated randomization schedule. After one year, the control group was offered participation in CoDE. The investigators, participants and community health worker were not blinded to group assignment. Recruitment materials, study protocol and consent documents were approved by University of Texas Health Science Center at Houston Institutional Review Board.

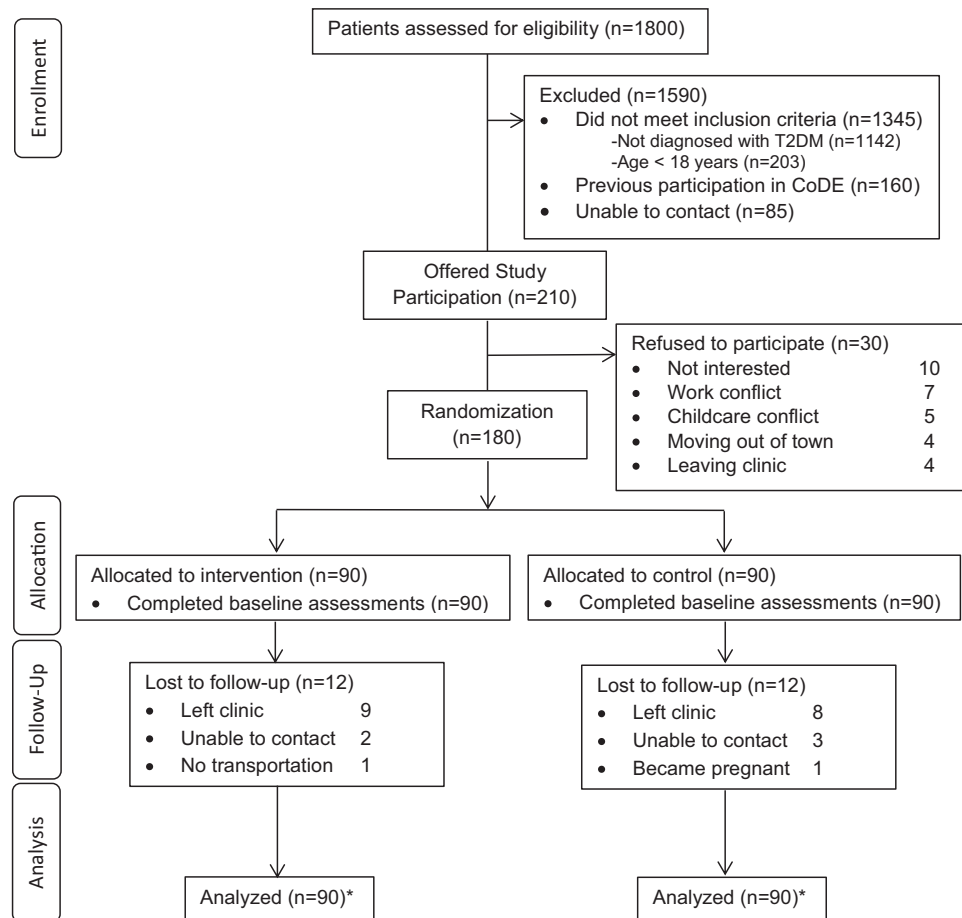


Fig. 1 – Study participant flow diagram of enrollment and loss to follow-up. *Intention-to-treat analysis.

2.2. Setting

The study site was located in an urban faith-based community health services clinic in Dallas, Texas which served exclusively uninsured patients of predominately Mexican American origin. Detailed information regarding the clinic staff and services has been previously reported [35]. In brief, the medical staff included three full-time physicians, bilingual medical assistants and clerical staff, but no nurses or CDEs. There was a Class D pharmacy on site to provide low cost medications to clinic clients, but no pharmacist, and by law the physicians were required to dispense medications to patients.

2.3. Study sample

The sample size was determined to be sufficient based upon a power analysis incorporating the effect size for change in HbA1c observed in an uncontrolled feasibility study of CoDE [13]. After adjusting upwards by 20% to account for expected attrition the final sample size of 180 subjects was established. This would provide 80% power for detecting a 10% or greater change in HbA1c in the intervention versus the control group with a type 1 error rate of 5%. Eligible patients were required to be active clinic clients; 18–75 years of age; diagnosed with type 2 diabetes mellitus (T2DM) treated with oral agents, insulin or

no medications; to have no advanced complications associated with diabetes; and to not be pregnant. The level of HbA1c was not used as a selection criterion because the intention of this study was to evaluate performance of the CHW under real life conditions and we relied upon randomization to distribute the ranges of HbA1c to both groups. Of 1800 potentially eligible patients, 75% did not meet inclusion criteria, 9% previously participated in the CoDE program, and 5% could not be contacted (Fig. 1). Of those subjects invited, 86% accepted study participation. Baseline assessments were completed by 100% of study subjects. Participants were enrolled from April 2005 through July 2005 followed by the 12-month study period.

2.4. Description of intervention and control conditions

Development and implementation of the CoDE protocol, adapted from expert resources, has been previously described [35–37]. Social cognitive theory guided the development of this protocol which places an emphasis on knowledge acquisition and the development of strategies for dealing with specific situations related to diabetes management [38]. Briefly, the bilingual CoDE health worker was an adult female lifelong member of the local Mexican American community, with a high school equivalency (General Educational Development:

GED) and certification from the State of Texas as a CHW. Her training to perform the role of diabetes educator and manager included 27 h of instruction in the local community from CDEs, Registered dietitians, and an endocrinologist. Competency of the CoDE CHW was formally assessed by written examination and clinical observation by the community diabetes instructors prior to the initiation of this clinical trial [35]. Per protocol, subjects in the intervention group received 7 h of contact with the CHW during scheduled appointments over 12 months in a private dedicated office space [35]. All sessions were conducted in the preferred language of the patient (English or Spanish). Printed educational materials were targeted for low literacy levels and available in either language. The three educational modules were delivered during individual 1 h sessions over the first 8 weeks. These sessions covered areas recommended by the American Diabetes Association including self-monitoring of blood glucose, 24 h recall diet history and individualized culturally appropriate meal planning, medication use, sick day rules, smoking cessation, exercise recommendations and information regarding long-term diabetes complications [39]. Blood glucose monitors and testing strips were provided free of charge. Thereafter, participants visited individually with the CHW by appointment during 1 h each quarter for reinforcement of skills and review of knowledge, and troubleshooting. The CHW facilitated immediate physician contact to address acute problems, assisted with pharmacy refills, and arranged specialty visits such as dental care and dilated retinal exams. The intervention group was followed by clinic physicians for usual medical care.

The control group received usual medical care at the discretion of the clinic physicians for the first 12 months. Subjects in this group were provided with a blood glucose monitor and testing strips free of charge and instructed in correct use of the device by medical assistants. Culturally tailored printed diabetes education materials were provided by physicians and clinic staff. All participants were given token cash incentives in the amount of \$5 during the third quarter and \$10 at the end of the study.

2.5. Study measures

Demographic information was collected at baseline. Clinical measures HbA1c, blood pressure, height and weight were collected at baseline and quarterly $\pm$ 2 weeks for 12 months. HbA1c was measured using a Bayer DCA 2000 + Analyzer[®], rigorously maintained for quality control. Blood pressure was determined as an average of three readings obtained 10 min apart using a digital sphygmomanometer by trained clinic staff. Height measurements were performed using a stadiometer, weight measurements were performed on a digital scale, and BMI was calculated using the standard formula (kg/m^2). Lipid status was assessed at baseline and 12 months at a laboratory compliant with regulatory standards. The number of unique antidiabetic medications at baseline was determined from patient history and pharmacy records. Medication changes, defined as modifications to patient drug regimens of the dosage of current therapeutic agents and/or the number of pharmacologic agents within each drug classification (antidiabetic, antihypertensive and lipid-lowering agents), were

collected quarterly from computerized pharmacy records for each study subject.

2.6. Statistical analysis

Data were collected from CoDE, clinic charts and pharmacy records by a trained research assistant. A database was constructed using double entry to insure accuracy. We compared baseline demographic and clinical characteristics using independent samples t-tests for continuous variables and Pearson χ^2 for categorical variables. The number of antidiabetic medications was calculated as a summation of one unit for each unique medication prescribed in this drug classification upon study entry. The number of changes made to the antidiabetic, antihypertensive and lipid-lowering regimens was calculated as a summation of every one unit change in dosage and/or pharmacological agent per quarter at all available time points over the study period. Differences between groups in diabetes medications at baseline were assessed using independent samples t-tests for continuous variables.

We analyzed repeated measures HbA1c, blood pressures, BMI and lipid values from study subjects as part of their original random group assignment following intention-to-treat principles. To detect differences in changes in outcomes over time between the intervention and control groups, linear mixed-models including a group-by-time interaction term and a random intercept were used. The mixed-model methodology used maximum likelihood estimation to obtain the parameter results. As all available data for every study subject were included in the analyses, missing values were not imputed.

All linear mixed-models initially included age, race and gender as well as covariates selected by clinical relevance as predictors or as potential confounders of the outcome. The most parsimonious final model for each repeated measures outcome was determined based upon Akaike Information Criterion (AIC) and likelihood ratio testing within nested models [40]. The change in HbA1c over time was further evaluated using a model which excluded baseline HbA1c from the repeated measures outcome of interest and used baseline HbA1c only as a covariate. Intervention effects comparing the adjusted mean changes in outcomes over the study period between intervention and control groups were calculated. We used Stata Statistical Software: Release 12 (StataCorp, College Station, TX) for all analyses [41].

3. Results

Baseline demographic and clinical characteristics for intervention and control groups are presented in Table 1. The majority of subjects were married non-smoking sedentary Mexican American middle-aged women with T2DM for less than 5 years and poor glycemic control ($\text{HbA1c} \geq 8\%$). The range for HbA1c was 5.3–14.5% for the intervention group and 5–17.3% for the control group. More than 20% of participants had less than a 6th grade education. The two groups did not differ with respect to the baseline clinical measures HbA1c, systolic and diastolic blood pressures, lipid levels (LDL, HDL,

Table 1 – Baseline demographic and clinical characteristics of study participants.

Characteristic	Intervention group (n = 90)	Control group (n = 90)	p-Value
HbA1c (%): mean $\pm$ S.D. ^a	8.9 $\pm$ 2.2	8.7 $\pm$ 2.3	0.53
HbA1c < 7%: n (%) ^b	23 (25.6)	22 (24.4)	0.86
HbA1c $\geq$ 8%: n (%) ^b	55 (61.1)	53 (58.9)	0.76
SBP (mm Hg): mean $\pm$ S.D. ^a	127 $\pm$ 20	125 $\pm$ 22	0.54
DBP (mm Hg): mean $\pm$ S.D. ^a	79 $\pm$ 11	79 $\pm$ 12	0.74
BMI (kg/m ²): mean $\pm$ S.D. ^a	32.7 $\pm$ 7.77	33.9 $\pm$ 8.24	0.28
LDL (mg/dL): mean $\pm$ S.D. ^a	110 $\pm$ 32.5	111 $\pm$ 34.2	0.89
HDL (mg/dL): mean $\pm$ S.D. ^a	46.6 $\pm$ 8.9	47.6 $\pm$ 14.5	0.57
Trig (mg/dL): mean $\pm$ S.D. ^a	229 $\pm$ 266	247 $\pm$ 233	0.62
Age (years): mean $\pm$ S.D. ^a	47.9 $\pm$ 10.99	45.7 $\pm$ 10.69	0.18
Number DM meds at baseline: mean $\pm$ S.D. ^a	1.27 $\pm$ 0.75	1.24 $\pm$ 0.72	0.78
Diabetes duration (years): mean $\pm$ S.D. ^a	4.8 $\pm$ 4.59	4.54 $\pm$ 5.64	0.73
Race: n (%) ^b			
Hispanic	73 (81.1)	74 (82.2)	0.97
African American	11 (12.2)	10 (11.1)	
Other	6 (6.7)	6 (6.7)	
Gender: n (%) ^b			
Male	30 (33.3)	41 (45.6)	0.09
Female	60 (66.7)	49 (54.4)	
Education level: n (%) ^b			
<6 years	21 (24.1)	20 (23.8)	0.40
6–11 years	40 (46)	46 (54.8)	
$\geq$ 12 years	26 (29.9)	18 (21.4)	
Marital status: n (%) ^b			
Never married	20 (22.2)	19 (21.4)	0.72
Married	52 (57.8)	56 (62.9)	
Divorced/separated/widowed	18 (20)	14 (15.7)	
Working status: n (%) ^b			
Working	28 (31.1)	43 (47.8)	0.06
Unemployed	52 (57.8)	37 (41.1)	
Disabled/retired	10 (11.1)	10 (11.1)	
Country of birth: n (%) ^b			
United States	20 (22.2)	27 (30)	0.31
Other	70 (77.8)	63 (70)	
Level of activity: n (%) ^b			
None	43 (48.9)	56 (62.2)	0.20
1–4 times (run/walk 1 mile in past month)	23 (26.1)	17 (18.9)	
>4 times (run/walk 1 mile in past month)	22 (25)	17 (18.9)	
Smoking status: n (%) ^b			
Never	55 (61.1)	45 (50)	0.13
Former smoker	26 (28.9)	27 (30)	
Current smoker	9 (10)	18 (20)	
Alcohol use: n (%) ^b			
Never	62 (68.9)	54 (60)	0.45
Former user	2 (2.2)	2 (2.2)	
Current user	26 (28.9)	34 (37.8)	

Note: DM: diabetes mellitus; BMI: body mass index; LDL: low density lipoprotein; HDL: high density lipoprotein; Trig: triglycerides; SBP: systolic blood pressure; DBP: diastolic blood pressure.

^a Independent samples t-test.

^b Pearson χ^2 .

triglycerides) or BMI. Of the 90 patients who completed the baseline assessment in the intervention group, 92% attended all 7 CoDE program sessions (data not shown). At study entry there was no difference between the groups in the number of antidiabetic medications prescribed (Table 1). There was no significant difference between the groups with respect to the

number of changes made to the antidiabetic, antihypertensive or lipid-lowering medication regimens over the study period (data not shown).

A total of 156 participants completed the 12 months study (14.4% attrition) with subject losses to follow-up evenly distributed between the two groups (Fig. 1). Both groups

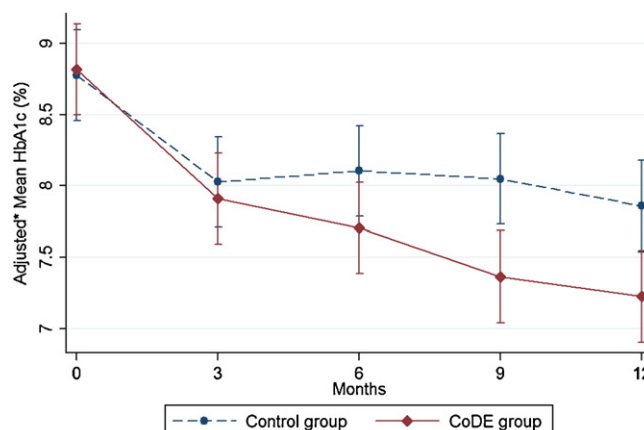


Fig. 2 – Change in HbA1c over 12 months with 95%CI.

experienced a decrease in HbA1c during the first quarter, but only the intervention group showed ongoing improvement for the duration of the study (Fig. 2).

Adjusted mean changes in HbA1c are presented in Table 2. The difference in mean changes of HbA1c over the study period showed a statistically significant intervention effect ($-.7\%$, $p = .02$) for the CoDE intervention group compared with the control group. The CoDE group achieved a greater reduction in HbA1c (-1.6% , $p < .001$) than the control group ($-.9\%$, $p < .001$). The results of the linear mixed-model for HbA1c which included baseline HbA1c only as a covariate are presented in Fig. 3. This model also included covariates to adjust for duration of T2DM, number of diabetes medications prescribed at baseline, and the number of changes made to the antidiabetic medication regimen over the study period. The group-by-time interaction was statistically significant for the third and fourth quarters indicating that the CoDE intervention group exhibited greater improvement over time during the second half of the study. All covariates were statistically significant as detractors to HbA1c improvement. For every 1% increase in baseline HbA1c the HbA1c at the end of the study was increased by $.25\%$ ($p < .001$). For every one year increase in the duration of diabetes the final HbA1c was increased by $.05\%$ ($p < .05$). Every one unit increase in the number of diabetes medications taken at baseline increased the final HbA1c by $.42\%$ ($p < .05$) and every one unit change made to the diabetes medication regimen over the study period increased the final HbA1c by $.4\%$ ($p < .001$).

Secondary outcomes are shown in Table 2. The control group experienced a statistically significant decrease in LDL (-9.8 mg/dL, $p < .01$). No other significant changes were observed within groups or between groups for blood pressures, BMI or lipid status. Group assignment was not statistically significant in any of the secondary outcomes analyses. The number of changes made to antihypertensive and lipid lowering medications were statistically significant covariates in the repeated measures systolic blood pressure model and lipid models respectively (data not shown). No adverse effects were noted among study participants as a result of the intervention.

4. Discussion

In this randomized controlled trial of uninsured adults with T2DM the average decrease in HbA1c levels from baseline to 12 months follow-up was 0.7% greater for the CoDE intervention group than for the usual care control group. This significant difference provides evidence for the effectiveness of the culturally tailored CHW-led CoDE intervention in improving glycemic control in Mexican Americans. Our study joins the limited body of literature that has employed rigorous methodology to conduct studies that compared health outcomes of patients receiving CHW-led DSME with usual care control groups. This study also demonstrates that it is possible to conduct a robust clinical trial within a community setting in an exclusively uninsured patient population. To understand the marked decline of HbA1c observed during the first quarter in both intervention and control groups we considered the Hawthorne Effect, which is a demonstrable benefit from simply participating in a clinical trial, as well as the initiation of or modifications to diabetes pharmacological therapy during the first quarter as potential explanations for this HbA1c reduction [42]. We felt it was essential to control for this time period by excluding baseline HbA1c from the repeated measures and instead used baseline HbA1c as a covariate. Medical management for diabetes can have a profound effect on HbA1c levels, and details about medication use are rarely reported by studies involving CHW-led DSME interventions [17]. To overcome this limitation we included the number of diabetes medications reported at baseline and the number of changes made to the diabetes medication regimen of each subject over the course of the study in our repeated measures model. Patients who were taking diabetes medications at baseline and in whom medication adjustments were made during the course of the study experienced less improvement in glycemic control. We are not aware of other randomized controlled trials of culturally tailored DSME interventions in socially disadvantaged populations that have included both of these important confounders when analyzing changes observed for HbA1c. In our study, a clinically meaningful

Table 2 – Adjusted mean changes in clinical measurements for study participants.^b

	Changes from baseline to 12 months within groups								Difference in mean change	
	Intervention group				Control group				Intervention vs. control group	
	Baseline	12 months	Change	N ^a	Baseline	12 months	Change	N ^a	Intervention effect	p
HbA1c, % (95%CI)	8.8(8.5, 9.1)	7.2(6.9, 7.5)	−1.6(−2.0, −1.2) [*]	78	8.8(8.5, 9.1)	7.9(7.5, 8.2)	−0.9(−1.3, −.54) [*]	78	−0.7(−1.2, −.1)	.02
SBP, mm Hg (95%CI)	127 (123, 131)	128(124, 132)	1.0(−3.1, 5.0)	76	125(121, 129)	128(124, 132)	3.3(−.7, 7.3)	78	−2.4(−8.3,3)	.42
DBP, mm Hg (95%CI)	80(77, 82)	80(77, 82)	−0.04(−2.9, 2.8)	76	78(75, 80)	78(76, 80)	0.02(−2.8, 2.8)	78	−.06(−4.3,9)	.98
BMI, kg/m ² (95%CI)	32.2(30.5, 33.9)	32.6(31, 34.3)	0.4(−.3, 1.1)	78	34.4(32.8, 36)	35.0(33.3, 36.7)	0.6(−.1, 1.3)	78	−.2(−1.1, .8)	.78
LDL, mg/dL (95%CI)	109(102, 115)	107(101, 113)	−1.7(−8.3, 5)	67	111(104, 118)	101(94, 108)	−9.8(−16.9, −2.8) ^{**}	62	8.2(−1.5, 17.8)	.10
HDL, mg/dL (95%CI)	45.4(42.9, 47.9)	46.0(43.5, 48.6)	0.6(−1.2, 2.5)	78	47.7(45.2, 50.3)	47.0(44.5, 49.5)	−0.7(−2.6, 1.2)	78	1.4(−1.3, 4)	.32
Trig, mg/dL (95%CI)	203(170, 236)	185(152, 219)	−17.5(−51.2,16.2)	78	237(204, 270)	234(201, 267)	−3.4(−37.2, 30.5)	78	−14.1(−61.9, 33.7)	.56

Note: HbA1c adjusted for baseline HbA1c, duration of diabetes, number of diabetes medications at baseline, and number of changes to diabetes medication regimen over 12 months. SBP: systolic blood pressure; DBP: diastolic blood pressure. SBP and DBP adjusted for age, race, gender, BMI at 12 months, activity level at baseline, alcohol use, smoking status, and total number of changes to antihypertensive medication over 12 months. BMI: body mass index; adjusted for gender, HbA1c change over 12 months, total number of changes to diabetes medication over 12 months, smoking status and activity level at baseline. LDL: low density lipoprotein; HDL: high density lipoprotein; Trig: triglycerides. LDL, HDL and Trig adjusted for gender, total number of changes to lipid lowering agents over 12 months, HbA1c change over 12 months, and change in BMI over 12 months.

^a Number with complete data at baseline and 12 month follow-up.

^b Linear mixed-models including group x time interaction and covariates.

^{*} $p < .001$.

^{**} $p < .01$.

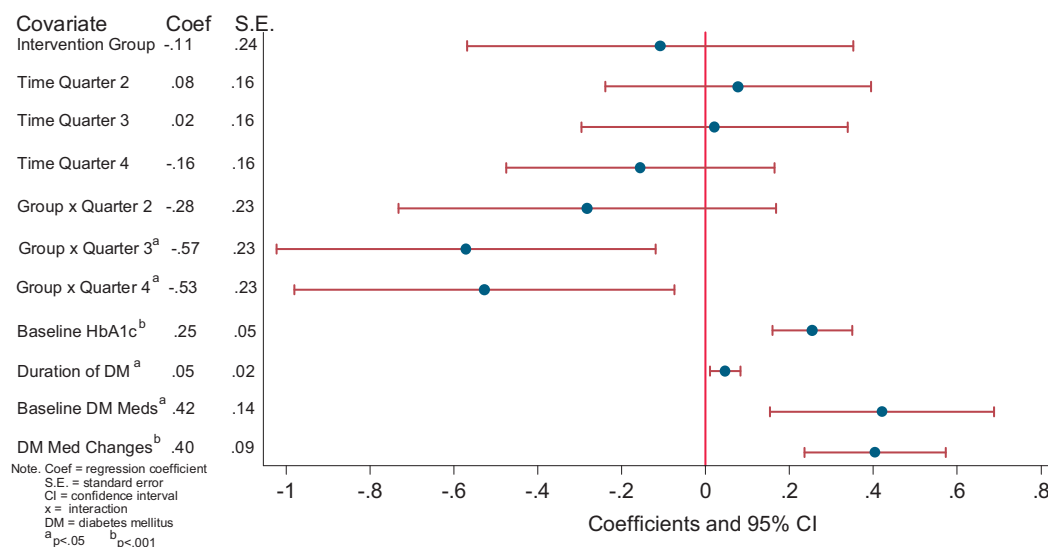


Fig. 3 – Linear mixed-model for HbA1c repeated measures.

intervention effect was observed for HbA1c even as we adjusted for the pharmacologic manipulation that was taking place in both groups through usual medical care. This highlights the importance of controlling for pharmacologic management when interpreting the health outcomes results achieved through DSME interventions. In addition, statistically significant differences between the groups did not become evident until the third and fourth quarters, supporting the recommendation that diabetes education interventions be of sufficient duration (≥ 6 months) [23].

The lack of significant differences in blood pressures over time was most likely due to the fact that participants were normotensive at baseline. Similarly, lipid levels on average were close to recommended target ranges so there was limited room for improvement [37]. The number of changes made to the antihypertensive and lipid-lowering medication regimens of study subjects contributed significantly to the blood pressure and lipid outcomes, while group assignment did not. Once again, it is essential to consider pharmacologic management when evaluating the health outcomes associated with DSME interventions. No significant impact on BMI was observed in the intervention group which is consistent with similar studies [8,10,12,15–17,43]. The failure of CoDE to reduce BMI can be explained in part by the fact that the primary focus of CoDE was not weight loss nor did it incorporate any specific exercise component due to lack of facilities and funding.

We contribute to the small group of studies which define specific primary roles for CHWs in diabetes management as integral members of the health care team. To our knowledge, this is the first study to report results from a one year office-based individual diabetes education and case management intervention which utilized a CHW in the capacity of sole provider of these services in the absence of more highly qualified supervisory personnel such as CDEs or nurses. The CoDE intervention differs from other culturally tailored DSME programs in similar disadvantaged populations in that CoDE is of relatively low intensity, with only 7 h of contact over 12 months [8,10,14,15,17–19,32,33,44–47]. Though this level of

intensity is not consistent with current recommendations for successful interventions, our intervention was effective in improving glycemic control potentially due to meaningful and productive office-based individual interaction with the CHW, high patient attendance at CoDE education and management sessions, and low attrition [23]. The fidelity of this intervention was high based upon the randomized design, established CHW and research assistant training, well-defined intervention protocols, quarterly data review and analysis, bi-monthly research team meetings, and weekly tracking of all study participants. Although both study groups received medical care within the same community clinic, the CHW functioned in an isolated, dedicated space that permitted interaction only with the intervention group, thereby minimizing contamination between the two study groups to the greatest extent possible.

Our findings support our main hypothesis that the CoDE program is an effective approach to reduce HbA1c levels with minimum investments in time and personnel. Interventions are often difficult to sustain after a study concludes, yet CoDE remains in operation at the original community clinic supported by philanthropic and local agency funding, and has been expanded to five other community sites who report encouraging preliminary data regarding CHW effectiveness as diabetes educators [48]. The benefits of diabetes education have been shown to decline within six months after an intervention ceases [9,20]. To address this issue, the CoDE program was designed from its inception to provide ongoing DSME after the first year as a maintenance program consisting of individual quarterly 1 h visits for follow-up HbA1c testing, review and updating of the meal plan, and ongoing reinforcement of skills and knowledge to maintain glycemic control [35].

This study has its limitations. The study groups could not be blinded to the physicians and this may in some way have influenced follow-up schedules of the control group and prompted more aggressive treatment of hypertension and lipid abnormalities. In addition, the small sample size may

have limited power to detect changes in outcomes other than HbA1c. The initial decrease in HbA1c observed in the control group may have been due in part to the fact that this group was also provided with glucose testing devices and testing strips. Furthermore, the various components of the CoDE program such as diabetes knowledge, meal planning and diet counseling, glucose monitoring, medication adherence, or case management could not be evaluated separately so it is not possible to state which components contributed the most to HbA1c improvement. As this study took place within a single community clinic where direct communication between the CHW and clinic physicians was possible in real time, the findings might not be generalizable to settings which do not have this mechanism in place. In addition, we cannot provide evidence that CoDE will have a similar impact on improvement of glycemic control in ethnic groups other than Mexican Americans.

This randomized controlled clinical trial has important implications for integrating CHWs as dedicated members of the health care team in underserved areas. Our goal is to promote development and expansion of CHW programs as one solution to narrow the widening gap in health disparities which challenges our health care system. The CoDE program needs to be further evaluated to fully understand which aspects of the intervention contribute the most to improvement in glycemic control and to assess the effectiveness of the management phase which continues after the first year. Further research is needed to clearly define the roles and responsibilities of CHWs in diabetes management and establish the effectiveness of CHWs in these roles in order for CHWs to become highly visible within the Chronic Care Model of evidence based practices [49].

Conflict of interest statement

The authors state that they have no conflicts of interest.

Acknowledgements

This study was supported by funding from the University of Texas School of Public Health and the Institute for Faith-Health Research-Dallas. We are grateful to Larry James, CEO of Central Dallas Ministries, James W. Walton, D.O., M.B.A. of Baylor Health Care System, Maura Thielen, M.D., Demetria M. Smith, M.D., Patricia L. Leczynski, M.D., Helen Rodriguez Farias, CHW, Armando Quiroz, CMA and Magdalena Lopez, CMA, CHW and the staff at Central Dallas Ministries Community Health Services clinic without whom this study would not have been possible.

REFERENCES

- [1] American Diabetes Association. Diabetes statistics. [http://www.diabetes.org/diabetes-basics/diabetes-statistics/?cr=redcpcggdiabetesstatisticsandinfo&gclid=CNvB2r7Q9LECFalgTAodC04ALw](http://www.diabetes.org/diabetes-basics/diabetes-statistics/?cr=redcpcggdiabetesstatisticsandinfo&gclid=CNvB2r7Q9LECFalgTAodC04ALw;); 2012 [accessed 22.08.12].
- [2] The United States Census Bureau. 2010 census briefs: the Hispanic population: 2010. <http://www.census.gov/prod/cen2010/briefs/c2010br-04.pdf>; 2012 [accessed 22.08.12].
- [3] U.S. Department of Health and Human Services. The Office of Minority Health: diabetes data/statistics. <http://minorityhealth.hhs.gov/templates/browse.aspx?lvl=3&lvlid=62>; 2012 [accessed 22.08.12].
- [4] Centers for Disease Control and Prevention. Diabetes data & trends. <http://www.cdc.gov/diabetes/statistics/esrd/fig5.htm>; 2012 [accessed 22.08.12].
- [5] The Diabetes Control and Complications Trial Research Group. The effect of intensive treatment of diabetes on the development and progression of long-term complications in insulin-dependent diabetes mellitus. *N Engl J Med* 1993;14:977–86.
- [6] UK Prospective Diabetes Study (UKPDS) Group. Intensive blood-glucose control with sulphonylureas or insulin compared with conventional treatment and risk of complications in patients with type 2 diabetes (UKPDS 33). *Lancet* 1998;9131:837–53.
- [7] American Diabetes Association. Executive summary: standards of medical care in diabetes – 2012. *Diabetes Care* 2012;34–10.
- [8] Brown SA, Garcia AA, Kouzekanani K, Hanis CL. Culturally competent diabetes self-management education for Mexican Americans: the Starr County border health initiative. *Diabetes Care* 2002;25:259–68.
- [9] Norris SL, Chowdhury FM, Van Le K, Horsley T, Brownstein JN, Zhang X, et al. Effectiveness of community health workers in the care of persons with diabetes. *Diabet Med* 2006;5:544–56.
- [10] Babamoto KS, Sey KA, Camilleri AJ, Karlan VJ, Catalasan J, Morisky DE. Improving diabetes care and health measures among hispanics using community health workers: results from a randomized controlled trial. *Health Educ Behav* 2009;1:113–26.
- [11] Valen MS, Narayan S, Wedeking L. An innovative approach to diabetes education for a Hispanic population utilizing community health workers. *J Cult Divers* 2012;1:10–7.
- [12] Brown SA, Garcia AA, Winter M, Silva L, Brown A, Hanis CL. Integrating education, group support, and case management for diabetic Hispanics. *Ethn Dis* 2011;1:20–6.
- [13] Culica D, Walton JW, Harker K, Prezio EA. Effectiveness of a community health worker as sole diabetes educator: comparison of CoDE with similar culturally appropriate interventions. *J Health Care Poor Underserved* 2008;4:1076–95.
- [14] Castillo A, Giachello A, Bates R, Concha J, Ramirez V, Sanchez C, et al. Community-based Diabetes Education for Latinos: the Diabetes Empowerment Education Program. *Diabetes Educ* 2010;4:586–94.
- [15] Lorig K, Ritter PL, Villa FJ, Armas J. Community-based peer-led diabetes self-management: a randomized trial. *Diabetes Educ* 2009;4:641–51.
- [16] Philis-Tsimikas A, Fortmann A, Lleba-Ocana L, Walker C, Gallo LC. Peer-led diabetes education programs in high-risk Mexican Americans improve glycemic control compared with standard approaches: a Project Dulce promotora randomized trial. *Diabetes Care* 2011;9:1926–31.
- [17] Spencer M, Rosland A, Kieffer E, Sinco B, Valerio M, Palmisano G, et al. Effectiveness of a community health worker intervention among African American and Latino adults with type 2 diabetes: a randomized controlled trial. *Am J Public Health* 2011;12:2253–60.
- [18] Rosal MC, Ockene IS, Restrepo A, White MJ, Borg A, Olendzki B, et al. Randomized trial of a literacy-sensitive, culturally tailored diabetes self-management intervention for low-income Latinos: Latinos en control. *Diabetes Care* 2011;4:838–44.

- [19] Sixta C, Ostwald S. Texas-Mexico border intervention by promotores for patients with type 2 diabetes. *Diabetes Educ* 2008;34:299–309.
- [20] Nam S, Janson SL, Stotts NA, Chesla C, Kroon L. Effect of culturally tailored diabetes education in ethnic minorities with type 2 diabetes: a meta-analysis. *J Cardiovasc Nurs* 2011.
- [21] Hunt CW, Grant JS, Appel SJ. An integrative review of community health advisors in type 2 diabetes. *J Community Health* 2011;5:883–93.
- [22] Witmer A, Seifer SD, Finocchio L, Leslie J, O'Neil EH. Community health workers: integral members of the health care work force. *Am J Public Health* 1995;8(Pt (1)):1055–8.
- [23] Glazier RH, Bajcar J, Kennie NR, Willson K. A systematic review of interventions to improve diabetes care in socially disadvantaged populations. *Diabetes Care* 2006;7:1675–88.
- [24] Nemcek MA, Sabatier R. State of evaluation: community health workers. *Public Health Nurs* 2003;4:260–70.
- [25] Cherrington A, Ayala GX, Amick H, Scarinci I, Allison J, Corbie-Smith G. Applying the community health worker model to diabetes management: using mixed methods to assess implementation and effectiveness. *J Health Care Poor Underserved* 2008;4:1044–59.
- [26] American Association of Diabetes Educators. AADE position statement. Community health workers in diabetes management and prevention. *Diabetes Educ* 2009;48S:52S.
- [27] American Public Health Association. Policy Statement Database: Support for community health workers to increase health access and reduce health inequities. <http://www.apha.org/advocacy/policy/policysearch/default.htm?id=1393;2012> [accessed 22.08.12] Policy Number 20091.
- [28] Keller T, Borges WJ, Hoke MM, Radasa T. Promotores and the Chronic Care Model: an organizational assessment. *J Community Health Nurs* 2011;2:70–80.
- [29] Ayala G, Vaz L, Earp J, Elder J, Cherrington A. Outcome effectiveness of the lay health advisor model among Latinos in the United States: an examination by role. *Health Educ Res* 2010;25:815–40.
- [30] Nelson K, Chapko K, Reiber G, Boyko E. The association between health insurance coverage and diabetes care: data from the 2000 Behavioral Risk Factor Surveillance System. *Health Serv Res* 2005;40:361–72.
- [31] Centers for Disease Control and Prevention. Diabetes data and trends: age-adjusted rates of ever attended diabetes self-management class per 100 adults with diabetes, United States, 2000–2006. http://www.cdc.gov/diabetes/statistics/preventive/fY_class.htm; 2012 [accessed 23.08.12].
- [32] Gilmer T, Philis-Tsimikas A. Outcomes of Project Dulce: a culturally specific diabetes management program. *Ann Pharmacother* 2005;39:817–22.
- [33] Lujan J, Ostwald SK, Ortiz M. Promotora diabetes intervention for Mexican Americans. *Diabetes Educ* 2007;4:660–70.
- [34] Healthy Communities Institute. Healthy North Texas: Community Dashboard. <http://www.healthyntexas.org>; 2012 [accessed 22.08.12].
- [35] Culica D, Walton JW, Prezio EA. CoDE: Community Diabetes Education for uninsured Mexican Americans. *Proc (Bayl Univ Med Cent)* 2007;2:111–7.
- [36] Franz M, editor. A core curriculum for diabetes educators. 5th ed., Chicago, IL: American Association of Diabetes Educators; 2003.
- [37] American Diabetes Association. Standards of medical care in diabetes – 2012. *Diabetes* 2012;S11–63.
- [38] Bandura A. Social cognitive theory: an agentic perspective. *Annu Rev Psychol* 2001;1:26.
- [39] Funnell MM, Brown TL, Childs BP, Haas LB, Hosey GM, Jensen B, et al. National standards for diabetes self-management education. *Diabetes Care* 2012;S101–8.
- [40] Posada D, Buckley T. Model selection and model averaging in phylogenetics: advantages of akaike information criterion and Bayesian approaches over likelihood ratio tests. *Syst Biol* 2004;53:793–808.
- [41] StataCorp. Stata statistical software: release 12. College Station, TX: StataCorp, LP; 2011.
- [42] McCarney R, Warner J, Iliffe S, van Haselen R, Griffin M, Fisher P. The Hawthorne effect: a randomised, controlled trial. *BMC Med Res Methodol* 2007;30.
- [43] Rosal MC, White MJ, Restrepo A, Olendzki B, Scavron J, Sinagra E, et al. Design and methods for a randomized clinical trial of a diabetes self-management intervention for low-income Latinos: Latinos en control. *BMC Med Res Methodol* 2009;81.
- [44] Balcazar HG, Byrd TL, Ortiz M, Tondapu SR, Chavez M. A randomized community intervention to improve hypertension control among Mexican Americans: using the promotoras de salud community outreach model. *J Health Care Poor Underserved* 2009;4:1079–94.
- [45] Joshi C, Rangel L, Garcia O, Brownson C, O'Toole M. Integration of a promotora-led self-management program into a system of care. *Diabetes Educ* 2007;33:151S–8S.
- [46] Otero-Sabogal R, Arretz D, Siebold S, Hallen E, Lee R, Ketchel A, et al. Physician-community health worker partnering to support diabetes self-management in primary care. *Qual Prim Care* 2010;6:363–72.
- [47] Rothschild SK, Martin MA, Swider SM, Lynas CT, Avery EF, Janssen I, et al. The Mexican-American Trial of Community Health workers (MATCH): design and baseline characteristics of a randomized controlled trial testing a culturally tailored community diabetes self-management intervention. *Contemp Clin Trials* 2012;2:369–77.
- [48] Walton JW, Snead CA, Collinsworth AW, Schmidt KL. Reducing diabetes disparities through the implementation of a community health worker-led diabetes self-management education program. *Fam Community Health* 2012;2:161–71.
- [49] Wagner E, Austin B, Davis C, Hindmaarsch M, Schaefer J, Bonomi A. Improving chronic illness care: translating evidence into action. *Health Aff* 2001;20:64–78.